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Exploring socio-behavioural correlates of depression and insomnia among college students in Cuttack, Odisha

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Abstract

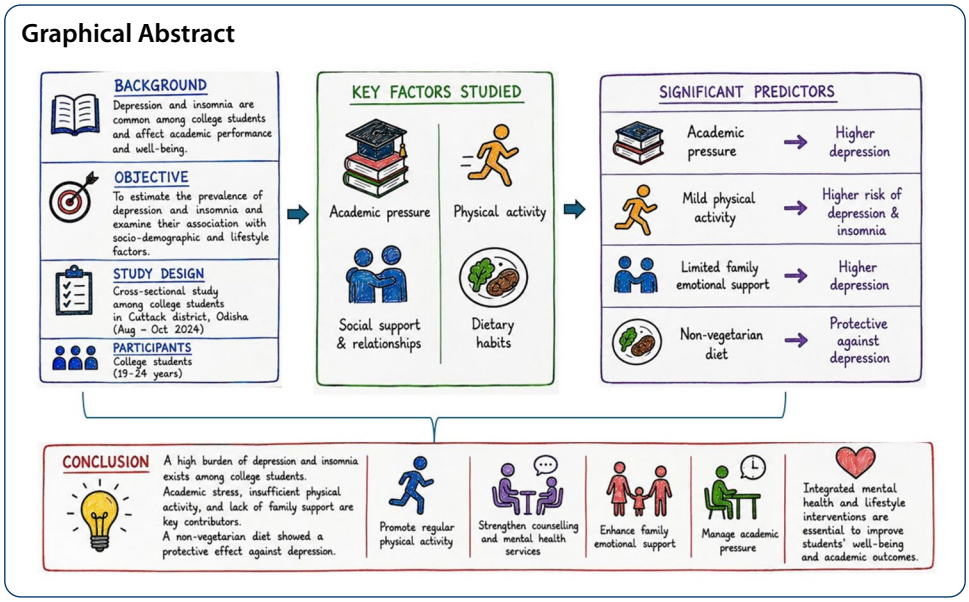
Depression and insomnia are more common mental health challenges for young adults, with an impact on academic performance and general health. However, there have been few studies on the interplay of socio-behavioural factors in college students in Eastern India. This study aimed to estimate the prevalence of depressive symptoms and insomnia and assess their relationship to socio-demographic and lifestyle factors among college students in Cuttack, Odisha. A cross-sectional survey was conducted from August to October 2024 among 432 students recruited through multistage sampling. The CES-D, Athens Insomnia Scale, and IPAQ-SF were used to measure depressive symptoms, insomnia, and physical activity, respectively. The data were analyzed using univariate and multivariate logistic regression. The prevalence of depressive symptoms (CES-D score ≥ 16) was 67.2%, while insomnia (35.8%) was significantly associated with very modest physical activity (AOR: 1.86, 95% CI 1.18–2.91) and depressive symptoms (AOR: 2.14, 95% CI 1.36–3.32). Academic pressure was also strongly associated with depressive symptoms (AOR: 3.09, 95% CI 1.49–6.33). A non-vegetarian diet was associated with lower odds of depressive symptoms (AOR: 0.47, 95% CI 0.27–0.77), and sharing problems with friends was associated with lower rates of depressive symptoms (AOR: 0.55, 95% CI 0.34–0.85). As a whole, the findings show a high prevalence of depressive symptoms and insomnia among college students. These findings highlight a need for institutional mental health support, access to counselling, regular physical activity, academic stress management, and enhanced emotional support to support students' health.

Keywords Depressive symptoms, Insomnia, College students, Physical activity, Academic stress, Social support, India

1 Introduction

Depression is a leading global cause of disability, impacting around 280 million people worldwide, as reported by the World Health Organisation. Mental health issues, especially among adolescents and young adults, significantly add to the global disease burden, with depression being a leading cause of illness, disability, and suicide. Additionally,

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sleep disorders like insomnia are rising as significant public health concerns, negatively affecting academic achievement, quality of life, and mental health. Recent estimates from the Global Burden of Disease show that depressive disorders continue to be among the top contributors to years lived with disability globally [1, 2].

In India, mental health issues pose a major public health concern. The National Mental Health Survey shows that nearly one in seven people will experience a mental disorder at some point, with depressive disorders notably contributing to disabilities among adolescents and young adults. The India State-Level Disease Burden Study highlights a growing burden of depression across states over the last thirty years. Among college students, depressive symptoms vary widely, affecting roughly 20% to over 70%, depending on the population, assessment methods, and setting. Additionally, insomnia and poor sleep quality are becoming more common among university students, driven by academic stress, lifestyle shifts, excessive screen use, and psychosocial pressures [3–8]. Nevertheless, data from Eastern India, especially Odisha, are scarce despite fast urban growth, shifting behavioural trends, and rising academic competition among youth.

Emerging evidence indicates that depression and insomnia often occur together and share many behavioural and psychosocial factors. Sleep issues can intensify depressive symptoms while depression can also deteriorate sleep quality, revealing a complex, bidirectional link. Past research highlights physical inactivity, academic stress, lack of social support, unhealthy eating habits, and excessive screen time as key contributors to poor mental health and sleep problems among college students [5–10].

College students are especially vulnerable because they face numerous academic, social, and developmental challenges during a crucial life stage. While several studies have independently explored depression or sleep issues among university students, few have examined both conditions together using validated screening tools, along with their shared socio-behavioural factors. Most existing research comes from northern and southern India, with limited data from Eastern India. Regional variations in culture, education, and lifestyle can impact the prevalence of depression and insomnia, making

it difficult to apply findings from other regions to Odisha. Hence, region-specific data is essential to develop targeted mental health programmes and public policies.

Unlike earlier research that looked at depression or insomnia in isolation, this study evaluates both conditions together using validated screening tools. It also identifies socio-behavioural factors that are common or unique to each condition among college students in Eastern India. This region-specific data can inform targeted mental health strategies and policy development.

The study specifically aimed to estimate the prevalence of depressive symptoms and insomnia among college students in Cuttack district, Odisha, and to analyse their shared and individual socio-behavioural associations using validated instruments.

2 Materials and methods

2.1 Study setting

The study was conducted in government and private degree colleges located in Cuttack district, Odisha, India. The participating institutions included colleges offering undergraduate and postgraduate programmes in Arts, Science, and Commerce streams.

2.2 Participants

Students aged 19–24 years were selected for this study.

2.3 Inclusion criteria

- Undergraduate and postgraduate students aged 19–24 years enrolled in the selected colleges during the study period.
- Students willing to participate and provide informed consent.

2.4 Exclusion criteria

- Students with self-reported chronic medical illnesses (e.g., neurological, endocrine, or other chronic conditions) or previously diagnosed psychiatric disorders that could influence depression or sleep status.

2.5 Sample size

Findings from previous literature show the prevalence of depression as 58.4% with 5% absolute precision and a 5% alpha level. The sample size was calculated using the formula for cross-sectional studies: $n = 4pq/d^2$ [9]. Considering a 10% non-response rate, we arrived at a final sample size of 432 subjects.

2.6 Sampling procedure

A multistage sampling technique involving stratified, cluster, and simple random sampling methods was employed. In the first stage, a comprehensive list of all government and private degree colleges in Cuttack district, Odisha, was prepared to serve as the sampling frame. In the second stage, colleges were stratified by management (government/private), and eligible colleges were selected. In the third stage, the Arts, Science, and Commerce streams were chosen because they represent the major academic disciplines with the highest student enrolment in the district. Departments within the selected colleges were considered as clusters. In the final stage, students were selected

by simple random sampling from the enrolment registers of the selected departments. Only students who met the eligibility criteria and provided written informed consent were included in the study.

2.7 Variable selection

Independent variables were selected in advance based on existing research, biological reasoning, and a conceptual model outlining factors associated with depression and insomnia among college students. Variables such as socio-demographic, behavioural, psychosocial, and family-related factors identified in earlier studies as potential confounders were included. The selection process was driven mainly by scientific evidence rather than by statistical significance alone.

2.8 Data collection tools

A pre-designed questionnaire was utilised to obtain socio-demographic details such as age, gender, number of siblings, maternal occupation, perceived financial status, academic stress, and paternal history of alcohol use. Standardised instruments were employed to assess physical activity, depression, and insomnia.

1. International Physical Activity Questionnaire – Short Form (IPAQ-SF)

   The IPAQ-SF was used to assess participants' physical activity over the previous 7 days. It captures information on time spent in walking, moderate, and vigorous activities, as well as sedentary behaviour. Based on the responses, total physical activity was calculated in Metabolic Equivalent of Task (MET)-minutes per week. Participants were categorised into mild (< 600 MET-min/week), moderate (600–3000 MET-min/week), and high (> 3000 MET-min/week) activity levels. The tool has demonstrated acceptable internal consistency, with a reported Cronbach's alpha of 0.758 [11].

2. Centre for Epidemiologic Studies Depression Scale (CES-D)

   Depressive symptoms were assessed using the CES-D scale, which consists of 20 items reflecting experiences during the past week. Each item is scored on a four-point Likert scale ranging from 0 (rarely) to 3 (most of the time), resulting in a total score between 0 and 60. A score of 16 or above was considered indicative of depressive symptoms. The scale has high internal consistency (Cronbach's alpha ≈ 0.90) and established validity [12].

3. Athens Insomnia Scale (AIS)

   The AIS was used to measure insomnia symptoms. It is a self-administered tool comprising eight items that assess sleep-related difficulties and their daytime consequences. Each item is rated on a scale from 0 (no problem) to 3 (severe problem), yielding a total score ranging from 0 to 24. A score ≥ 6 was considered suggestive of insomnia [13].

The questionnaire was translated into Odia and subsequently back-translated into English to ensure linguistic accuracy. A consensus version was developed through discussions with clinical psychologists and public health experts. Pre-testing was conducted

among 20 students to identify potential issues, and necessary modifications were made. The finalised Odia version was used for data collection. Data were collected during scheduled visits to selected colleges following permission from the relevant authorities. Trained investigators administered questionnaires in classroom settings. Participants were briefed on the study's aims and reassured about confidentiality and voluntary participation before completing the questionnaires. The self-administered questionnaires were filled out anonymously to reduce interviewer and social desirability biases. Investigators were available to clarify any questions without affecting responses. Completed questionnaires were checked for completeness and consistency upon collection. Along with the standardised tools, factors like academic pressure, food preferences (vegetarian or non-vegetarian), strength of friendships (fair or strong), social media use as a favourite pastime (yes/no), and preferred person for sharing problems (family or friends) were evaluated. These variables were identified through a comprehensive literature review and expert input. The questionnaire was translated into Odia, the local language, using a forward–backward translation method. Content validity was verified by specialists in clinical psychology and public health. A pilot test with 20 college students was conducted to assess clarity, understanding, and cultural appropriateness. Based on their feedback, minor modifications were made before finalising.

2.9 Data collection procedure

Data were collected between August and October 2024 following approval from the participating colleges. Trained investigators visited the selected institutions during regular class hours and explained the objectives and procedures of the study to eligible students. Data collection did not coincide with the university examination period / coincided with the regular academic session), thereby addressing the potential influence of examination-related stress on the study outcomes After obtaining written informed consent, participants completed the anonymous self-administered questionnaire in the classroom under investigator supervision. Investigators were available to clarify queries without influencing participants' responses. Completed questionnaires were checked immediately for completeness and consistency before data entry.

2.10 Bias

Several measures were adopted to minimise potential sources of bias. Anonymous self-administered questionnaires were used to reduce social desirability bias, while standardized instructions were provided to all participants to minimise information bias. Investigators received uniform training before data collection to ensure consistency in questionnaire administration. The use of validated instruments further reduced measurement bias. Questionnaires with incomplete information on outcome variables or key covariates were excluded from the final analysis.

2.11 Statistical analysis

Data were entered and analysed using Epi Info version 7.0 (Centers for Disease Control and Prevention, Atlanta, USA). Categorical variables were summarised using frequencies and percentages. Univariate logistic regression analysis was performed to examine associations between socio-demographic and lifestyle-related variables and depressive symptoms or insomnia. Variables included in the multivariable logistic regression models

were selected based on existing literature, biological plausibility, and the conceptual framework. Adjusted prevalence odds ratios (Adjusted PORs) with 95% confidence intervals (CIs) were reported. Statistical significance was considered at $p < 0.05$. Missing data were minimal (< 5%), and complete-case analysis was performed after excluding questionnaires with incomplete outcome or key covariate information.

2.12 Research ethics

Ethical clearance was obtained from the Institutional Ethics Committee of SCB Medical College and Hospital, Cuttack (Approval No. 1829 dated 24 July 2024). Written informed consent was obtained from all participants before enrolment in the study.

3 Results

A total of 432 study subjects were recruited for the study to identify the prevalence of depression and insomnia and their association with social and lifestyle factors. The mean age of the participants was 19.6 with a standard deviation of 1.5 years. Table 1 shows that the majority of respondents were ≤ 20 years of age (70.6%), Female (65.5%), staying outside the hostel (50.2%), and had a maximum of two siblings (80.3%), and only 19.7% had more than 2 siblings. The mothers of most of them were unemployed/homemakers (78.7%), and they had a positive self-perception of their financial condition (67.8%). Additionally, 16.2% of respondents reported experiencing academic pressure, and 13% revealed that their fathers consumed alcohol. Around 67.2% of participants screened positive for depressive symptoms (CES-D score ≥ 16), and 35.8% for insomnia (AIS score ≥ 6).

Table 2 shows the association between lifestyle and social factors and insomnia among participants. Significant associations were observed for dietary habits and physical activity.

Participants with mild physical activity had higher odds of insomnia *(POR* = 1.99, 95% CI 1.29–3.09; $p = 0.002$) than those reporting moderate or high physical activity.

Table 1 Socio-demographic profile of the respondents

Parameters	Categories	Frequency	Percent
Age	≤ 20	305	70.6
	≥ 20	127	29.4
Gender	Female	283	65.5
	Male	149	34.5
Mother's Job Profile	Employed	92	21.3
	Unemployed	340	78.7
Staying in Hostel	No	215	49.8
	Yes	217	50.2
No of siblings	> 2	85	19.7
	0–2	347	80.3
Self-Perception about financial condition	Good	293	67.8
	Poor	139	32.1
Academic pressure	Yes	70	16.2
	No	362	83.8
Alcohol abuse in father	Yes	56	13
	No	376	87
Physical Activity	Mild	285	66
	Moderate/high	147	34

Table 2 Association of lifestyle and social factors with insomnia among participants

Predictors	Categories	Insomnia				*P*-value	Prevalence odds ratio (POR)
		Yes		No			
		Count	%	Count	%		
Academics pressure	Yes	30	42.9%	40	57.1%	0.220	1.42 (0.845–2.39)
	No	125	34.5%	237	65.5%		
Alcohol abuse in father	Yes	19	33.9%	37	66.1%	0.880	0.91 (0.50–1.64)
	No	136	36.2%	240	63.8%		
Food habit	Non-veg	98	32.8%	201	67.2%	**0.05***	0.65 (0.427–0.99)
	Veg	57	42.9%	76	57.1%		
Physical activity	Mild	117	41.1%	168	58.9%	**0.002***	1.99 (1.29–3.09)
	Moderate/high	38	25.9%	109	74.1%		
Using social media is your favourite pastime	No	136	35.1%	251	64.9%	0.412	0.74 (0.396–1.39)
	Yes	19	42.2%	26	57.8%		
Relationship with friends	Fair	48	36.4%	84	63.6%	0.914	1.03 (0.67–1.58)
	Strong	107	35.7%	193	64.3%		
Share your problems with	Friends	73	32.9%	149	67.1%	0.193	0.76 (0.516–1.13)
	Family	82	39.0%	128	61.0%		

Reference category (Ref.): Academic pressure (No), Alcohol abuse in father (No), Food habit (Vegetarian), Physical activity (Moderate/High), Social media as favourite pastime (Yes), Relationship with friends (Strong), Share your problems with (Family). Associations were examined using univariate logistic regression

POR: Prevalence Odds Ratio; CI: Confidence Interval

$p < 0.05$ was considered statistically significant; bold and *indicates values are significant

Table 3 Association of lifestyle and social factors with depressive symptoms in respondent

Predictors	Categories	Depression				*P*-value	Prevalence odds ratio (POR)
		Mild		Moderate/Severe			
		Count	%	Count	%		
Academics pressure	Yes	59	84.3%	11	15.7%	**0.001***	3.04 (1.54–5.99)
	No	231	63.8%	131	36.2%		
Alcohol abuse in father	Yes	38	67.9%	18	32.1%	1.000	1.03 (0.57–1.89)
	No	252	67.0%	124	33.0%		
Food habit	Non-veg	183	61.2%	116	38.8%	**0.001***	0.38 (0.23–0.624)
	Veg	107	80.5%	26	19.5%		
Physical activity	Mild	211	74.0%	74	26.0%	0.001*	2.45 (1.62–3.73)
	Moderate /high	79	53.7%	68	46.3%		
Using social media is your favourite pastime	No	254	65.6%	133	34.4%	0.064	0.477 (0.223–1.02)
	Yes	36	80.0%	9	20.0%		
Relationship with friends	Fair	84	63.6%	48	36.4%	0.318	0.799 (0.519–1.228)
	Strong	206	68.7%	94	31.3%		
Share your problems with	Friends	132	59.5%	90	40.5%	**0.001***	0.48 (0.32–0.729)
	Family	158	75.2%	52	24.8%		

Bold and *indicates values are significant

Table 3 shows the relationship between socio-behavioural factors and depression among respondents. Significant links were found for academic pressure, dietary habits, physical activity, and preferred emotional support sources. Students experiencing academic pressure had increased odds of depressive symptoms ($POR = 3.04$, 95% CI 1.54–5.99; $p = 0.001$). Engaging in mild physical activity was also linked to higher odds of depressive symptoms ($POR = 2.45$, 95% CI 1.62–3.73, $p = 0.001$). Additionally, dietary patterns ($POR = 0.38$, 95% CI 0.23–0.62, $p = 0.001$) and preferred emotional support

Table 4 Multivariate logistic regression showing association between risk factors and Insomnia and depression

Risk factor	Category	Insomnia adjusted POR (95% CI)	*p*-value	Depressive symptoms adjusted POR (95% CI)	*p*-value
Academic pressure	No (Ref.)	Reference	–	Reference	–
	Yes	1.43 (0.82–2.46)	0.203	3.09 (1.49–6.33)	0.002
Alcohol abuse in father	No (Ref.)	Reference	–	Reference	–
	Yes	0.92 (0.47–1.73)	0.779	1.26 (0.61–2.57)	0.527
Food habit	Vegetarian (Ref.)	Reference	–	Reference	–
	Non-vegetarian	0.74 (0.47–1.14)	0.178	0.47 (0.27–0.77)	0.004
Physical activity	Moderate/High (Ref.)	Reference	–	Reference	–
	Mild	1.86 (1.18–2.91)	0.007	2.14 (1.36–3.32)	0.001
Social media as a favourite pastime	Yes (Ref.)	Reference	–	Reference	–
	No	0.82 (0.43–1.55)	0.530	0.47 (0.20–1.05)	0.066
Relationship with friends	Strong (Ref.)	Reference	–	Reference	–
	Fair	0.98 (0.62–1.52)	0.911	0.62 (0.37–1.01)	0.055
Preferred person to share problems with	Family (Ref.)	Reference	–	Reference	–
	Friends	0.88 (0.57–1.31)	0.516	0.55 (0.34–0.85)	0.008

Values are adjusted prevalence odds ratios obtained from multivariable logistic regression analysis. Reference categories are indicated as "(Ref.)" in the Category column

Adjusted POR: Adjusted Prevalence Odds Ratio; CI: Confidence Interval; Ref.: Reference category

Statistically significant associations ($p < 0.05$) are shown in bold

sources (*POR* = 0.48, 95% *CI* 0.32–0.73, $p = 0.001$) showed significant associations with depressive symptoms.

Table 4 illustrates that, in the multivariable analysis, Mild physical activity remained significantly associated with insomnia (Adjusted *POR* = 1.86, 95% CI 1.18–2.91; $p = 0.007$). Regarding depressive symptoms factors such as academic pressure (*Adjusted POR* 3.09, 95% *CI* 1.49–6.33), mild physical activity (*Adjusted POR* = 2.14, 95% *CI* 1.36–3.32), dietary pattern (*Adjusted POR* = 0.47, 95% *CI* 0.27–0.77), and the preferred source of emotional support (*Adjusted POR* = 0.55, 95% *CI* 0.34–0.85) remained significantl associated after adjusting for potential confounders.

4 Discussion

We found a high prevalence of depressive symptoms (67.2%) and insomnia (35.8%) among college students in Cuttack, Odisha, which is a significant mental health burden in this population. Mild physical activity was the only common factor independently associated with depressive symptoms and insomnia. Academic pressure, dietary habits, and emotional support were also associated with depressive symptoms, while paternal alcohol use, friendship quality, and social media use were not independently associated with either outcome after adjustment. These findings underscore the role of modifiable behavioural and psychosocial factors in student mental health. While 13% of the participants reported paternal alcohol abuse, no significant association was found with depressive symptoms or insomnia. This was contrary to findings from Hebbani et al., who reported a relationship between parental alcoholism and psychological distress in college students [14], for example. Factors in the study population, family functioning, and alcohol exposure severity and assessment method might be important. Further studies of family functioning and alcohol use at the level of parents to get a better sense of the relation are required. Mild physical activity was found to be significantly associated

with both depressive symptoms and insomnia, the only common predictor in our study. This is corroborated by previous studies on physical activity, depressive symptoms, and sleep quality among college students [15-17]. Nearly two-thirds of participants reported only mild physical activity, and we need to increase the opportunities for regular physical activity in educational settings. The integration of structured physical activity and lifestyle programmes into college health initiatives may help both psychological well-being and sleep health. Dietary habits were strongly linked with depressive symptoms in students, with a non-vegetarian diet showing a lower risk of depressive symptoms. This should be taken with care, as it was only vegetarian eating or non-vegetarian eating, and no assessment of dietary quality/nutrient content, meal frequency, or food diversity was made. The association may have been a reflection of larger cultural, socioeconomic, or lifestyle differences and not of a direct protective effect of non-vegetarian food consumption. There are inconsistent relationships between vegetarian diets and depressive symptoms observed in previous studies [18-20]. Further dietary assessments are needed in future research. Academic pressure was one of the strongest independent predictors of depressive symptoms but was not significantly associated with insomnia after adjustment. Academic demands, fear of failure, parental expectations, and career concerns may contribute to psychological distress among college students. Liu et al. [21] have found similar results. These findings suggest the need to integrate academic stress management, counselling, and psychological support within higher education. Emotional support also had a strong association with depressive symptoms. Students who preferred to share their problems with friends had a lower risk for depressive symptoms than those who preferred to share with family members. This result should be taken in a cautious manner, as the measure is about the preferred source of problem sharing, not about the quality or adequacy of family support. Future research using validated measures of perceived social support and family functioning should be done. In this case, no significant associations were found between social media use and friendship quality and depressive symptoms or insomnia as a result of adjustment. However, social media use was assessed only as a preferred pastime and did not include duration, frequency, night-time use, or problematic use. Friendship quality was assessed in a relatively simple way. Future studies should use more sophisticated measures such as digital media use and social support to more comprehensively evaluate digital media behaviour[22-23]. Overall, mild physical activity, academic pressure, dietary habits and emotional support were shown to be important socio-behavioural factors for mental health in this group. These potentially modifiable factors point to the need for institutional mental health care, counselling services, regular physical activity, academic stress management and supportive family and social environments. Such integrated approaches may enhance psychological well-being and healthier sleep among college students.

5 Limitations

Limitations: There are several limitations to this study. The cross-sectional design inhibits causal inference, and some socio-behavioural determinants were assessed using simple self-reported measures. The presence of confounding from unmeasured factors cannot be ruled out, and the direct relationship between depressive symptoms and insomnia was not investigated. While multistage sampling was employed, the sample size calculation did not take into account the design effect, which may have

underestimated the sample size. Finally, the findings may have been quite limited in generalisation beyond college students in Cuttack, Odisha. Future studies should employ longitudinal designs, make use of appropriate sampling adjustments and investigate barriers to psychological help-seeking, including stigma, awareness and access to counselling services.

6 Future research

Future studies should utilize longitudinal designs to understand temporal links between socio-behavioural factors, depression and insomnia. It should be possible to identify barriers to psychological help-seeking for college students, such as stigma, awareness, and accessibility of counselling services, to improve timely and effective mental health interventions.

7 Conclusion

The study found a high prevalence of depressive symptoms (67.2%) and insomnia (35.8%) among college students in Cuttack, Odisha. Mild physical activity was associated with both outcomes, and academic pressure, dietary habits, and emotional support were all highly associated with depressive symptoms. These results underscore the urgent need for institutional mental health support, counselling services, regular physical activity, academic stress management, and greater family involvement to promote student well-being. Since cross-sectional work is so common in such populations, we need longitudinal studies to understand the relationships and identify barriers to psychological help-seeking among college students.

8 Recommendations

Academic institutions should offer counselling services and carry out regular mental health screening with standardised tools in conjunction with psychology professionals. Students should be encouraged to take part in mental health awareness programmes, regular physical activity, and academic stress management programmes. Family and emotional support can be further enhanced for students. Longitudinal studies of socio-behavioural factors, dietary patterns, depression, and.

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Sai Chandan Das: Writing – original draft, Project administration, Methodology, Investigation, Data curation, Conceptualization. Deep Shikha: Writing – original draft, Project administration, Methodology, Investigation, Data curation, Conceptualization; Prabhudarsan Saho: Writing – review & editing, Formal analysis. Prasanta Kumar Sethi: Writing – review & editing. Mausumi Mishra: Writing – review & editing, Resources, Methodology, Conceptualization. R V Manasa: Writing – review & editing, Formal analysis, Data curation; Bindu Naik: Writing – review & editing, Formal analysis; V. Kumar:Writing – review & editing, Formal analysis;

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Data availability
The datasets generated and/or analysed during the current study are available from Dr. Deep Sikha and Dr. Sai Chandan Das on reasonable request. Requests for access to the data should be addressed to Dr. Sai Chandan Das, Department of Community Medicine, Kalinga Institute of Medical Sciences, KIIT University, Bhubaneswar, Odisha, India, Email: saichandan.das@kims.ac.in and Dr. Deep Sikha, Himalayan Institute of Medical Sciences, Swami Rama Himalayan University, Dehradun, Uttarakhand, India, Email: deepshikha@srhu.edu.in.

Declarations

Ethics approval and consent to participate
The study was conducted in accordance with the ethical standards of the institutional ethics committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. Ethical approval for the study was obtained from the Institutional Ethics Committee of SCB Medical College & Hospital, Cuttack, Approval No.1829. Informed consent was obtained from all individual participants included in the study.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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